

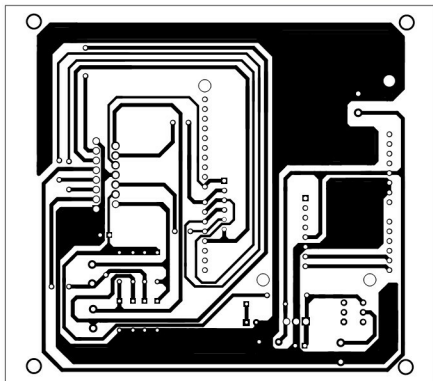


Fig. 3: Actual-size PCB layout of two-wheel self-balancing robot

Arduino IDE, compile the sketch and upload it to the board.

Construction and testing

An actual-size PCB layout of the two-wheel self-balancing robot is shown in Fig. 3 and its components layout in Fig. 4. Use a suitable metallic chassis for the robot. Mount the Arduino board on the chassis of the robot. Then place the PCB on top of the Arduino board.

First, burn the `mpu_calibration.ino` code into Arduino, attach the MPU-6050 and upload the sketch to the Arduino board. Then open the serial monitor and select baud rate as 115200. After initialisation of the DMP (digital motion processing)—on-board processor of the MPU-6050—press 'a' followed by Enter key. You will get a screen as shown in Fig. 5. Note that your offset values will be different.

Note down the offset values. Then open the main `two_wheel.ino` code and enter the test offset values. Compile and upload the main code to the

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